

Common Fixed Points of Strongly Compatible Mappings in Modular G -Metric Spaces

Abstract

We introduce an orbital strengthening of ω -compatibility for pairs of self-maps on modular G -metric spaces. The new notion, called *strong ω -compatibility of orbital type*, combines asymptotic commutation of compositions along coincidence sequences with a convergence-transfer requirement for those compositions. This work investigates common fixed point theorems in the context of G -complete modular G -metric spaces under Jungck-type modular G -contractions. Under a standard range-inclusion hypothesis, we provide existence and uniqueness proofs that do not require continuous mappings. Beyond the primary theorem, we derive specialized cases, including a Banach-type principle for single maps and results for iterated contractions. We also demonstrate that the framework remains robust when completeness is limited to the range of the transformation. The practical importance of these findings is finally confirmed through an application to nonlinear integral equations

MSC (2020): 47H10, 54H25, 46A80

Keywords: Modular G -metric space; strong ω -compatibility; Jungck contraction; common fixed point; orbital convergence

1 Introduction

Fixed point theory in generalized distance structures has developed into a major theme of nonlinear analysis, with applications to functional and integral equations as well as iterative schemes. The idea behind G -metric spaces by Mustafa and Sims [4] gave birth to a well-defined three-variable

framework, allowing for the natural extension of various classical fixed-point arguments. Parallel to this development, Chistyakov [2] introduced metric modular spaces to formalize parameter-dependent, distance-like quantities, a direction that closely conforms with the study of modular function spaces.

By integrating these two paradigms, Azadifar et al. [1] defined the modular G -metric space. This hybrid structure has since become a fertile area for research, leading to numerous fixed-point and common fixed-point results established under diverse contractive regimes (see, for example, [1, 6, 7]).

Central to the evolution of common fixed-point theory is the requirement of a "compatibility" mechanism between mappings. While Jungck [3] initially framed compatibility as a convergence-based commutation property within standard metric spaces, these concepts have been extensively adapted. In the context of G metric, compatible and weakly compatible mappings are well-documented [5], and more recently several iterations of ω -compatibility have emerged within the modular literature of G metric.

A persistent technical challenge in these settings, even in classical frameworks, is the reliance on auxiliary regularity conditions. Common fixed-point theorems for a pair (S, T) under Jungck-type contractions often necessitate the continuity of at least one operator, the closeness of a range, or specific orbital conditions to facilitate the limit-passing process in compositions and to transition from coincidence points to common fixed points. This paper addresses this limitation by proposing a novel compatibility condition within the modular G metric setting.

We do so by introducing strong ω -compatibility of orbital type: besides asymptotic commutation along coincidence sequences, the definition incorporates a direct convergence-transfer requirement for the compositions. This specific mechanism is engineered to circumvent traditional continuity hypotheses while preserving the analytical rigor essential for establishing the existence and uniqueness of solutions. Beyond these abstract theoretical developments, we substantiate the utility of the proposed framework by applying it to a nonlinear integral equation. This application highlights how the modular G -metric approach provides a robust foundation for resolving operator equations frequently encountered in functional analysis.

The manuscript is arranged as follows: **Section 2** reviews fundamental concepts and establishes the auxiliary lemmas necessary for the subsequent proofs. **Section 3** introduces the notion of *strong ω -compatibility of orbital type* and explores its theoretical significance. Our primary results and their corresponding corollaries are presented in **Section 4**, followed by an

illustrative numerical example in **Section 5**. Finally, **Section 6** details the application of our findings to a nonlinear integral equation, and concluding remarks are offered in **Section 7**.

2 Preliminaries

In this section, we review the foundational concepts of modular G -metric spaces, primarily following the axiomatic framework founded by Azadifar *et al.* [1].

Definition 2.1 (Modular G -metric [1]). if we let X be a non-empty set. A mapping

$$\omega_G : (0, \infty) \times X \times X \times X \rightarrow [0, \infty],$$

denoted by $\omega_{G,\lambda}(x, y, z)$, is called a *modular G -metric* on X if for all $x, y, z, a \in X$ and $\lambda, \mu > 0$, the following conditions hold:

- (G1) $\omega_{G,\lambda}(x, y, z) = 0$ for all $\lambda > 0$ if and only if $x = y = z$;
- (G2) $\omega_{G,\lambda}(x, x, y) > 0$ for all $\lambda > 0$ whenever $x \neq y$;
- (G3) $\omega_{G,\lambda}(x, x, y) \leq \omega_{G,\lambda}(x, y, z)$ for all $\lambda > 0$ whenever $y \neq z$;
- (G4) $\omega_{G,\lambda}$ is symmetric under any permutation of the variables x, y , and z ;
- (G5) (*rectangle inequality*) $\omega_{\lambda+\mu}^G(x, y, z) \leq \omega_\lambda^G(x, a, a) + \omega_\mu^G(a, y, z)$.

The pair (X, ω^G) is called a *modular G -metric space*.

Definition 2.2 (Modular G -convergence and Cauchy property [1]). Let (X, ω^G) be a modular G -metric space and (x_n) a sequence in X .

- (i) We say that x_n *modular G -converges* to $x \in X$, written $x_n \rightarrow x$, if

$$\omega_\lambda^G(x_n, x, x) \rightarrow 0 \quad \text{as } n \rightarrow \infty, \text{ for all } \lambda > 0.$$

- (ii) We say that (x_n) is *modular G -Cauchy* if for every $\varepsilon > 0$ and every $\lambda > 0$ there exists N such that

$$\omega_\lambda^G(x_n, x_m, x_\ell) < \varepsilon \quad \text{for all } n, m, \ell \geq N.$$

- (iii) The space is G -complete if every modular G -Cauchy sequence is modular G -convergent in X .

Proposition 2.3 (Equivalent convergence forms [1]). *Let (X, ω^G) be a modular G -metric space and (x_n) a sequence in X . The following are equivalent:*

- (a) $x_n \rightarrow x$;
- (b) $\omega_\lambda^G(x_n, x, x) \rightarrow 0$ for all $\lambda > 0$;
- (c) $\omega_\lambda^G(x_n, x_n, x) \rightarrow 0$ for all $\lambda > 0$;
- (d) $\omega_\lambda^G(x_n, x_m, x) \rightarrow 0$ as $n, m \rightarrow \infty$ for all $\lambda > 0$.

Remark 2.4. Limits in a modular G -metric space are unique. Indeed, if $x_n \rightarrow x$ and $x_n \rightarrow y$, then by Proposition 2.3 and the rectangle inequality one gets

$$\omega_\lambda^G(x, y, y) = 0 \quad \text{for all } \lambda > 0,$$

hence $x = y$.

Lemma 2.5 (Two-sequence transfer). *Let (X, ω^G) be a modular G -metric space. Suppose (a_n) and (b_n) satisfy*

$$\omega_\lambda^G(a_n, a_n, b_n) \rightarrow 0 \quad \text{for all } \lambda > 0,$$

and $a_n \rightarrow a$. Then $b_n \rightarrow a$.

Proof. Setting $\lambda > 0$. By the rectangle inequality with $x = b_n$, $y = a$, $z = a$ and $a = a_n$, and using symmetry,

$$\omega_{2\lambda}^G(b_n, a, a) \leq \omega_\lambda^G(b_n, a_n, a_n) + \omega_\lambda^G(a_n, a, a) = \omega_\lambda^G(a_n, a_n, b_n) + \omega_\lambda^G(a_n, a, a).$$

The right-hand side tends to 0, hence $\omega_{2\lambda}^G(b_n, a, a) \rightarrow 0$. By Proposition 2.3, $b_n \rightarrow a$. $\square$

Lemma 2.6 (Geometric decay implies Cauchy). *Let (X, ω^G) be a modular G -metric space and (y_n) a sequence in X . Assume that for each fixed $\lambda > 0$ there exist constants $C_\lambda \geq 0$ and $q \in (0, 1)$ such that*

$$\omega_\lambda^G(y_n, y_n, y_{n+1}) \leq C_\lambda q^n \quad \text{for all } n \geq 0.$$

Then (y_n) is modular G -Cauchy.

Proof. Let $\varepsilon > 0$ and $\lambda > 0$. Choose N such that $\sum_{j=N}^{\infty} C_{\lambda} q^j < \varepsilon$. Let $m > n \geq N$. Repeated use of the rectangle inequality yields

$$\omega_{(m-n)\lambda}^G(y_n, y_n, y_m) \leq \sum_{j=n}^{m-1} \omega_{\lambda}^G(y_j, y_j, y_{j+1}) \leq \sum_{j=n}^{m-1} C_{\lambda} q^j \leq \sum_{j=N}^{\infty} C_{\lambda} q^j < \varepsilon.$$

Using symmetry and Proposition 2.3(d) (applied with parameter $(m-n)\lambda$ and then letting $m, n \rightarrow \infty$) gives the modular G -Cauchy property in the sense of Definition 2.2(ii). $\square$

Remark 2.7. Lemma 2.6 serves as the natural modular G -metric, which is classical in the term-cancellation argument used in standard metric spaces. Essentially, it establishes that ensuring geometric (or otherwise summable) control over successive distances is sufficient to satisfy the Cauchy criterion via iterative application of the rectangle inequality

3 Strong ω -Compatibility of Orbital Type

Definition 3.1 (Strong ω -compatibility (orbital type)). Let (X, ω^G) be a modular G -metric space and $S, T : X \rightarrow X$. We say that (S, T) is *strongly ω -compatible of orbital type* if for every sequence (x_n) in X and every $z \in X$ such that

$$Sx_n \rightarrow z \quad \text{and} \quad Tx_n \rightarrow z,$$

the following hold for all $\lambda > 0$:

(SC1) (*asymptotic commutation*) $\omega_{\lambda}^G(STx_n, TSx_n, TSx_n) \rightarrow 0$ and $\omega_{\lambda}^G(TSx_n, STx_n, STx_n) \rightarrow 0$;

(SC2) (*orbital convergence transfer*) $\omega_{\lambda}^G(STx_n, Sz, Sz) \rightarrow 0$ and $\omega_{\lambda}^G(TSx_n, Tz, Tz) \rightarrow 0$.

Remark 3.2. Condition (SC2) is the key strengthening relative to several versions of ω -compatibility used in the literature: it replaces the continuity-based procedure that is typically used to pass from $Sx_n \rightarrow z$ and $Tx_n \rightarrow z$ to conclusions about STx_n and TSx_n .

4 Main Results

Definition 4.1 (Jungck-type modular G -contraction). Let (X, ω^G) be a modular G -metric space. A pair (S, T) of self-maps is said to satisfy a *Jungck-type modular G -contraction* if there exists $k \in (0, 1)$ such that for all $x, y, z \in X$ and all $\lambda > 0$,

$$\omega_\lambda^G(Sx, Sy, Sz) \leq k \omega_\lambda^G(Tx, Ty, Tz). \quad (1)$$

Lemma 4.2 (Coincidence of limits under asymptotic commutation). *Let (X, ω^G) be a modular G -metric space, and let (a_n) , (b_n) be sequences in X such that for every $\lambda > 0$,*

$$\omega_\lambda^G(a_n, b_n, b_n) \rightarrow 0.$$

If $a_n \rightarrow a$ and $b_n \rightarrow b$ in the modular G -sense, then $a = b$.

Proof. By Lemma 2.5, from

$$\omega_\lambda^G(a_n, b_n, b_n) \rightarrow 0$$

and $b_n \rightarrow b$, it follows that $a_n \rightarrow b$. But we are also given that $a_n \rightarrow a$. By uniqueness of limits in modular G -metric spaces, we obtain $a = b$. $\square$

Theorem 4.3 (Common fixed point under strong ω -compatibility). *Let (X, ω^G) be a G -complete modular G -metric space. Let $S, T : X \rightarrow X$ satisfy:*

- (H1) $S(X) \subseteq T(X)$;*
- (H2) (S, T) is strongly ω -compatible of orbital type (Definition 3.1);*
- (H3) (S, T) satisfies the Jungck-type modular G -contraction (1) for some $k \in (0, 1)$.*

Then S and T have a unique common fixed point in X .

Step 1: Using Iterative by construction.we need to prove by the induction method Given an element x_n , we select x_{n+1} such that

$$Tx_{n+1} = Sx_n.$$

For clarity purposes in the subsequent analysis, we define the sequence $\{y_n\}_{n \geq 0}$ by setting

$$y_n := Tx_{n+1} = Sx_n.$$

Step 2: In this second step, we need to prove the geometric Decay of Successive Modular G -distances. Fix an arbitrary $\lambda > 0$. By applying the contractive condition (1) to the triplet

$$(x, y, z) = (x_n, x_n, x_{n+1}),$$

we obtain

$$\omega_{G,\lambda}(Sx_n, Sx_n, Sx_{n+1}) \leq k \omega_{G,\lambda}(Tx_n, Tx_n, Tx_{n+1}).$$

In terms of the sequence $\{y_n\}$, for any $n \geq 1$, this inequality takes the form

$$\omega_{G,\lambda}(y_n, y_n, y_{n+1}) \leq k \omega_{G,\lambda}(y_{n-1}, y_{n-1}, y_n).$$

Iterating yields

$$\omega_\lambda^G(y_n, y_n, y_{n+1}) \leq k^n \omega_\lambda^G(y_0, y_0, y_1) \quad (n \geq 0). \quad (2)$$

Using Cauchy property and existence of the limit, let $C_\lambda := \omega_\lambda^G(y_0, y_0, y_1)$ and $q := k$. Then (2) gives $\omega_\lambda^G(y_n, y_n, y_{n+1}) \leq C_\lambda q^n$ for all n . By Lemma 2.6, (y_n) is modular G -Cauchy, hence there exists $u \in X$ such that $y_n \rightarrow u$. Since $y_n = Sx_n$ and $y_{n-1} = Tx_n$, we have

$$Sx_n \rightarrow u \quad \text{and} \quad Tx_n \rightarrow u.$$

Examining the coincidence at the limit, we apply Definition 3.1 to (x_n) and $z = u$. From **(SC2)**,

$$STx_n \rightarrow Su \quad \text{and} \quad TSx_n \rightarrow Tu.$$

From **(SC1)**, for every $\lambda > 0$,

$$\omega_\lambda^G(STx_n, TSx_n, TSx_n) \rightarrow 0.$$

Applying Lemma 4.2 with $a_n = STx_n$ and $b_n = TSx_n$, we obtain

$$Su = Tu.$$

We now establish that u is indeed a common fixed point. Fix an arbitrary $\lambda > 0$ and define

$$a_n := \omega_{G,\lambda}(y_n, y_n, Su).$$

By evaluating the contractive condition (1) at the points

$$(x, y, z) = (x_n, x_n, u),$$

we obtain

$$\omega_{G,\lambda}(Sx_n, Sx_n, Su) \leq k \omega_{G,\lambda}(Tx_n, Tx_n, Tu).$$

Given the identifications $Sx_n = y_n$, $Tx_n = y_{n-1}$, and the coincidence condition $Tu = Su$, the inequality simplifies to

$$a_n \leq k a_{n-1} \quad \text{for all } n \geq 1.$$

By iteration, it follows that

$$a_n \leq k^n a_0.$$

Since $k \in (0, 1)$, we have $a_n \rightarrow 0$ as $n \rightarrow \infty$, which implies

$$\omega_{G,\lambda}(y_n, y_n, Su) \rightarrow 0.$$

Noting that $y_n \rightarrow u$, an application of Lemma 2.5 (setting $a_n = y_n$ and $b_n \equiv Su$) necessitates that $Su = u$. Consequently,

$$Tu = Su = u,$$

confirming that u is a common fixed point for the pair (S, T) .

Uniqueness. To verify uniqueness, suppose there exists another common fixed point $v \in X$ such that

$$Sv = Tv = v.$$

Applying the contraction (1) to the triplet

$$(x, y, z) = (u, u, v)$$

yields

$$\omega_{G,\lambda}(u, u, v) = \omega_{G,\lambda}(Su, Su, Sv) \leq k \omega_{G,\lambda}(Tu, Tu, Tv) = k \omega_{G,\lambda}(u, u, v).$$

Since $k < 1$, this inequality holds if and only if

$$\omega_{G,\lambda}(u, u, v) = 0 \quad \text{for all } \lambda > 0.$$

By the identity axiom (G1), we conclude that $u = v$, thereby establishing the uniqueness of the common fixed point. Since $k \in (0, 1)$, we have $\omega_\lambda^G(u, u, v) = 0$ for all $\lambda > 0$, hence $u = v$. $\square$

Remark 4.4. No continuity of S or T is used. The orbital transfer condition (SC2) replaces the usual continuity-based limit passage for STx_n and TSx_n .

4.1 Consequences

Corollary 4.5 (Single-map Banach principle). *Let (X, ω^G) be G -complete and let $S : X \rightarrow X$ satisfy, for some $k \in (0, 1)$,*

$$\omega_\lambda^G(Sx, Sy, Sz) \leq k \omega_\lambda^G(x, y, z) \quad \forall x, y, z \in X, \quad \forall \lambda > 0.$$

Then S has a unique fixed point in X .

Proof. Apply Theorem 4.3 with $T = \text{Id}$. Then $S(X) \subseteq X$ is automatic, and (1) reduces to the stated hypothesis. Moreover, for any sequence with $Sx_n \rightarrow z$ and $x_n \rightarrow z$ (i.e. $Tx_n \rightarrow z$), we have $STx_n = Sx_n$ and $TSx_n = Sx_n$, so (SC1)–(SC2) hold trivially. $\square$

Corollary 4.6 (Iterated Jungck contraction). *Assume (H1)–(H2) of Theorem 4.3. Suppose there exist $m \in \mathbb{N}$ and $k \in (0, 1)$ such that*

$$\omega_\lambda^G(S^m x, S^m y, S^m z) \leq k \omega_\lambda^G(T^m x, T^m y, T^m z) \quad \forall x, y, z \in X, \quad \forall \lambda > 0.$$

Then S and T have a unique common fixed point.

Proof. Apply Theorem 4.3 to (S^m, T^m) . The inclusion $S(X) \subseteq T(X)$ implies $S^m(X) \subseteq T^m(X)$. The strong ω -compatibility of orbital type transfers to (S^m, T^m) by repeated application of Definition 3.1 along the same coincidence sequences. Let u be the unique common fixed point of S^m and T^m . Then Su is also a fixed point of S^m , since $S^m(Su) = S(S^m u) = Su$; similarly Tu is a fixed point of T^m . By uniqueness, $Su = u$ and $Tu = u$. $\square$

Corollary 4.7 (Completeness of the range). *Let (X, ω^G) be a modular G -metric space. Assume $T(X)$ is G -complete as a subspace and that (H1)–(H3) of Theorem 4.3 hold. Then S and T have a unique common fixed point.*

Proof. In the proof of Theorem 4.3, the sequence (y_n) lies in $T(X)$ because $y_n = Tx_{n+1}$. Lemma 2.6 shows (y_n) is modular G -Cauchy, hence convergent in $T(X)$ by completeness of $T(X)$. The remainder of the argument is unchanged. $\square$

4.2 Further Remarks

Remark 4.8 (Novelty and nonemptiness of the hypothesis). Many common fixed point theorems in modular G -metric spaces combine a Jungck-type inclusion $S(X) \subseteq T(X)$ and a modular contractive inequality with an additional regularity device (e.g. continuity of one of the maps, closedness of a range, or weaker compatibility assumptions) to pass to limits in the compositions and upgrade coincidence to a common fixed point. Theorem 4.3 replaces that device by an orbital compatibility mechanism: strong ω -compatibility of orbital type forces asymptotic commutation *and* transfers convergence of the compositions to the corresponding limit images, thereby avoiding continuity assumptions.

Remark 4.9 (A sufficient condition for strong ω -compatibility). If S and T are modular G -continuous and weakly compatible (i.e. $Sx = Tx \Rightarrow STx = TSx$), then the orbital transfer (SC2) holds automatically, and asymptotic commutation (SC1) follows along coincidence sequences. Hence, in the presence of continuity, Definition 3.1 is satisfied in the standard regime considered in common fixed point arguments.

5 Example and Application

5.1 Example

Example 5.1. Let $X = [0, \infty)$ and define, for $\lambda > 0$,

$$\omega_\lambda^G(x, y, z) = \frac{1}{\lambda}(|x - y| + |y - z| + |z - x|).$$

Then (X, ω^G) is a modular G -metric space (symmetry is clear; the rectangle inequality follows from the triangle inequality on $\mathbb{R}$). Define $S, T : X \rightarrow X$ by

$$Sx = \frac{x}{4}, \quad Tx = \frac{x}{2}.$$

Then $S(X) \subseteq T(X)$ since for any $w \geq 0$ we have $w = Tx$ with $x = 2w$. Moreover, for all $x, y, z \in X$ and $\lambda > 0$,

$$\omega_\lambda^G(Sx, Sy, Sz) = \frac{1}{4} \omega_\lambda^G(x, y, z) \quad \text{and} \quad \omega_\lambda^G(Tx, Ty, Tz) = \frac{1}{2} \omega_\lambda^G(x, y, z),$$

hence

$$\omega_\lambda^G(Sx, Sy, Sz) = \frac{1}{2} \omega_\lambda^G(Tx, Ty, Tz),$$

so (1) holds with $k = \frac{1}{2}$.

Finally, if $Sx_n \rightarrow z$ and $Tx_n \rightarrow z$, then $Tx_n \rightarrow z$ implies $x_n \rightarrow 2z$ (in the usual sense, hence also in the modular G -sense), so $Sx_n \rightarrow \frac{2z}{4} = \frac{z}{2}$. Since also $Sx_n \rightarrow z$, we obtain $z = \frac{z}{2}$ and hence $z = 0$. Thus any coincidence sequence can only converge to 0, and for such a sequence,

$$STx_n = \frac{x_n}{8} \rightarrow 0 = S0, \quad TSx_n = \frac{x_n}{8} \rightarrow 0 = T0,$$

and asymptotic commutation holds identically since $STx_n = TSx_n$ for all n . Therefore (SC1)–(SC2) are satisfied, and Theorem 4.3 yields the unique common fixed point 0.

5.2 Application to a Nonlinear Integral Equation

In this section we illustrate Corollary 4.5 by applying it to the existence and uniqueness of solutions of a nonlinear integral equation.

Let $X = C([0, 1], \mathbb{R})$ be the Banach space of continuous real-valued functions on $[0, 1]$, equipped with the supremum norm $\|\cdot\|_\infty$. Define a modular G -metric on X by

$$\omega_\lambda^G(f, g, h) = \frac{1}{\lambda} (\|f - g\|_\infty + \|g - h\|_\infty + \|h - f\|_\infty), \quad \lambda > 0.$$

It is straightforward to verify that (X, ω^G) is a G -complete modular G -metric space.

Consider the nonlinear integral equation

$$u(t) = \int_0^1 K(t, s) F(s, u(s)) ds, \quad t \in [0, 1], \quad (3)$$

where the functions $K : [0, 1] \times [0, 1] \rightarrow \mathbb{R}$ and $F : [0, 1] \times \mathbb{R} \rightarrow \mathbb{R}$ satisfy:

1. K is continuous on $[0, 1] \times [0, 1]$;
2. F is continuous;
3. there exists $L > 0$ such that

$$|F(t, x) - F(t, y)| \leq L|x - y| \quad \text{for all } t \in [0, 1], \quad x, y \in \mathbb{R};$$

4.

$$M := \sup_{t \in [0, 1]} \int_0^1 |K(t, s)| ds < \infty.$$

Define an operator $S : X \rightarrow X$ by

$$(Su)(t) = \int_0^1 K(t, s) F(s, u(s)) ds.$$

We show that S is a modular G -contraction.

Let $f, g \in X$. Then for each $t \in [0, 1]$,

$$|Sf(t) - Sg(t)| \leq \int_0^1 |K(t, s)| |F(s, f(s)) - F(s, g(s))| ds.$$

Using the Lipschitz condition on F , we obtain

$$|Sf(t) - Sg(t)| \leq L \int_0^1 |K(t, s)| |f(s) - g(s)| ds \leq LM \|f - g\|_\infty.$$

Taking the supremum over t yields

$$\|Sf - Sg\|_\infty \leq LM \|f - g\|_\infty.$$

Consequently, for all $f, g, h \in X$ and all $\lambda > 0$,

$$\omega_\lambda^G(Sf, Sg, Sh) = \frac{1}{\lambda} (\|Sf - Sg\|_\infty + \|Sg - Sh\|_\infty + \|Sh - Sf\|_\infty) \leq LM \omega_\lambda^G(f, g, h).$$

If $LM < 1$, then S is a modular G -contraction in the sense of Corollary 4.5. Therefore S admits a unique fixed point $u \in X$.

Hence equation (3) possesses a unique continuous solution on $[0, 1]$.

6 Conclusion

We introduced the notion of strong ω -compatibility of orbital type for pairs of self-maps on modular G -metric spaces and established a Jungck-type common fixed point theorem under this framework without assuming continuity. The primary methodological innovation of this work lies in the **orbital convergence transfer mechanism**. By implementing this framework, we effectively circumvent the standard reliance on continuity-based limit arguments typically required to transition from coincidence points to common fixed points. This development offers a more versatile analytical path in spaces where operator regularity may be difficult to establish.

Furthermore, we have substantiated the abstract theory through its application to nonlinear integral equations. This illustrates that the modular G -metric environment is not merely a theoretical generalization, but a functional setting capable of accommodating complex operator equations encountered in functional analysis. Ultimately, these results underscore the adaptability of the framework and its potential for broader utility in resolving existence and uniqueness problems in non-classical metric structures. . Possible extensions include multivalued mappings, Geraghty-type modular G -contractions, and adaptations to ordered or graph-structured modular G -metric spaces, as well as further applications to nonlinear functional and differential equations.

References

- [1] B. Azadifar, M. Maramaei and G. Sadeghi, *On the modular G -metric spaces and fixed point theorems*, J. Nonlinear Sci. Appl. **6** (2013), 293–304.
- [2] V. V. Chistyakov, *Metric modular spaces I: Basic concepts*, Nonlinear Anal. Theory Methods Appl. **72** (2010), no. 1, 1–14.
- [3] G. Jungck, *Compatible mappings and common fixed points*, Int. J. Math. Math. Sci. **9** (1986), no. 4, 771–779.
- [4] Z. Mustafa and B. Sims, *A new approach to generalized metric spaces*, J. Nonlinear Convex Anal. **7** (2006), no. 2, 289–297.

- [5] Z. Mustafa, *Common fixed points of weakly compatible mappings in G -metric spaces*, Appl. Math. Sci. **6** (2012), no. 92, 4589–4600.
- [6] G. A. Okeke and D. Francis, *Fixed point theorems in modular G -metric spaces*, J. Inequal. Appl. **2021** (2021), Article 163.
- [7] G. A. Okeke and D. Francis, *Fixed point theorems for asymptotically T -regular mappings in preordered modular G -metric spaces applied to solving nonlinear integral equations*, J. Anal. **30** (2022), 501–545.